

01 Requirements Specification

RAMIFY OS v11.0.6 | Client Requirements Alignment

1. Scope

RAMIFY OS v11.0.6 is the expanded client demonstration artefact for KIT300 Project 12. It preserves the deterministic trust boundary while adding the shopping journey, agent policies, human review, signed receipts, proof pack and formal assessment material.

2. In scope

- Local synthetic product-trust demo
- LangGraph/Ollama/Llama interpretation for free-form text only
- Deterministic RESOLVE/RATIFY checks
- Five persona policies
- Human review and successor receipt
- Action Gate, basket and simulated checkout
- Receipt verification, tamper demo and proof-pack export

3. Out of scope

- Real purchasing, payments or supplier integration
- Real product certification or regulatory assurance
- LLM-generated trust verdicts
- Production key management, revocation or governance

4. Requirements traceability

ID	Requirement	Implementation evidence	Acceptance evidence
FR-01	Local browser prototype and shopping demo	FastAPI backend, separate HTML pages, Shop-first flow	Smoke tests for /shop, /tour, /demo and API routes
FR-02	AI-agent layer for natural-language interpretation	LangGraph + Ollama + Llama 3.1 adapter for free-form Shop text	live_llm_smoke.py on presentation laptop; deterministic fallback in automated tests
FR-03	Deterministic RESOLVE/RATIFY trust checks	engine.py, resolve primitives, ratify checks and precedence	Full unittest suite and five-primitive visual trace
FR-04	Five required scenarios	Approved, Expired evidence, Seller risk, Recall, Substitution in Demo Story Library	Scenario coverage API and test suite
FR-05	Human-readable and JSON receipt	Receipt Summary, Evidence Trail and Signed JSON tabs	Receipt verification and tamper tests
FR-06	Mandatory human review when required	Needs me queue and successor receipt model	Review API tests and UI workflow
FR-07	Controlled follow-on action	Action Gate, basket and checkout require current purchase authority	Cart and Action Gate regression tests
SEC-01	Tamper-evident receipt	Canonical SHA-256 + Ed25519 demo signer	Browser verify, CLI verifier and tamper demo
NFR-01	Demo reliability and performance	Exact selection bypasses LLM; fallback timeout; no manual AI toggle	100-run deterministic benchmark under 50 ms